HOMEWORK 1: 1.1-2.6

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1.1. Let $x = (1,2,3)^T$, $y = (y_1,y_2,y_3)^T$, $z = (4,2,1)^T$. Compute $2x$, $3y$, and $x + 2y - 3z$.

Proof. 1: $2x = 2(1,2,3)^T = (2,4,6)^T$
2: $3y = 3(y_1,y_2,y_3)^T = (3y_1,3y_2,3y_3)^T$
3: $x+2y-3z = (1,2,3)^T+2(y_1,y_2,y_3)^T-(12,6,3)^T = (2y_1-11,2y_2-4,2y_3)^T$ □

1.2. Which of the following sets (with natural addition and multiplcation by a scalar) are vector spaces? Justify your answer.

a) The set of all continuous functions on the interval $[0,1]$;
b) The set of all non-negative functions on the interval $[0,1]$;
c) The set of all polynomials of degree exactly n;
d) The set of all symmetric $n \times n$ matrices, i.e the set of matrices $A = \{a_{j,k}\}_{j,k=1}^n$ such that $A^T = A$.

Proof. A is a vector space. Given two distinct functions f and g, and some $x \in [0,1]$, we can define vector addition and scalar multiplication as such: $(f+g)(x) = f(x) + g(x)$; $(\alpha f)(x) = \alpha f(x)$, where $\alpha \in \mathbb{R}$.

B is not a vector space. The set of all non-negative functions fail to be a vector space when we choose a function f and multiply it by a scalar $\alpha < 0$.

C is not a vector space. Addititive inverses exist.

D is a vector space. Because of the symmetric properties, scalar multiplication and vector addition would also give symmetric matrices. The zero vector is symmetric. □

1.3. True or false:

a) Every vector space contains a zero vector;
b) A vector space can have more than one zero vector;
c) An $m \times n$ matrix has m rows and n columns;
d) If f and g are polynomials of degree n, then $f + g$ is also a polynomial of degree n;
e) If f and g are polynomials of degree at most n, then $f+g$ is also a polynomial of degree at most n

Proof. TFTFT □

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1.4. Prove that a zero vector 0 of a vector space V is unique.

Proof. To prove uniqueness, assume that there exists two zero vectors, 0 and $\bar{0}$, both belonging to the vector space V. By our axioms, we know if $x \in V$, then $x+0 = x+\bar{0} = x$. Now, using our axioms, we can derive that $x+\bar{0} = x+0 \to \bar{0} = 0$. This shows that our distinct zero vectors are equal, and thus the zero vector is unique. □

1.5. What matrix is the zero vector of the space $M_{2\times 3}$

Proof. $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ □

1.6. Prove that the additive inverse, defined in Axiom 4 of a vector space is unique.

Proof. To prove uniqueness, assume there exists two additive inverses in V, that being $-v$ and $\bar{v}$. Using our axioms, we may derive that $v + (-v) = v + (\bar{v}) = 0$. Now, using our axioms, we can derive that $v + (-v) = v + (\bar{v}) \to -v = \bar{v}$. This shows that the additive inverse is unique. □

1.7. Prove that $0v = 0$ for any vector $v \in V$.

Proof. $0v = 0 \to (0+0)v = 0v \to 0v + 0v = 0v \to 0v = 0$. □

1.8. Prove that for any vector v its additive inverse $-v$ is given by $(-1)v$.

Proof. We know that $v + (-v) = 0$ through our axioms. Now, we can represent this using the distributive property: $(1 + (-1))v = 0 \to 0v = 0$ which is proven by our axiom for vector spaces. □

2.1. Find a basis in the space of 3×2 matrices $M_{3\times 2}$.

Proof.

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

□

2.2. True or false:

a) Any set containing the zero vector is linearly dependent;
b) A basis must contain 0;
c) subsets of linearly dependent sets are linearly dependent;
d) subsets of linearly independent sets are linearly independent;
e) If $\alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n = 0$, then all scalars α_k are zero;

Proof. TFFTF □

2.3. Recall, that a matrix is called symmetric if $A^T = A$. Write down a basis in the space of symmetric 2×2 matrices. How many elements are in the basis?

Proof. $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is the format for these symmetric matrices. Now, using a linear combination to represent this: $a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ The vectors next to the coefficients are the elements of the basis. □

2.4. Write down a basis for the space of

a) 3×3 symmetric matrices;
b) $n \times n$ symmetric matrices;
c) $n \times n$ antisymmetric ($A^T = -A$) matrices;

Proof. A 3×3 symmetric matrix can be represented as $\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}$. So, the basis would be $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. For B, we know symmetric matrices entail $e_{ij} = e_{ji}$, where $a_i i$ can be any number. We can define that set to be $\{e_{ii} : 1 \leq i \leq n\}$ or $\sum_{i=1}^{n} e_{ii}$. Now we must deal with the subdiagonals, corners and such. We can do such with the following: $\sum_{i=1}^{n} \sum_{j=1}^{i-1} e_{ij}$ for the upper diagonal and $\sum_{j=1}^{n} \sum_{i=1}^{j-1} e_{ij}$. We can simply rewrite this as $e_{ij}+e_{ji}$, giving us the upper and lower diagonal. Thus, we can create the basis set to be $\{e_{ii} : 1 \leq i \leq n\} \cup \{e_{ij} + eji : 1 \leq i < j \leq n\}$. For C, anti-symmetric matrcies must have zeros along the diagonal. So, $\{e_{ii} = 0 : 1 \leq i \leq n\}$. Now for the negatives, we can simply make the upper triangle or the lower triangle the negative, so let's define that as $\{e_{ij} - e_{ji} : 1 \leq i < j \leq n\}$. So for anti-symmetric sets, $\{e_{ii} = 0 : 1 \leq i \leq n\} \cup \{e_{ij} - e_{ji} : 1 \leq i < j \leq n\}$ □

2.5. Let a system of vectors $v_1, v_2, \ldots, v_r$ be linearly independent but not generating. Show that it is possible to find a vector v_{r+1} such that the system $v_1, v_2, \ldots, v_r, v_{r+1}$ is linearly independent.

Proof. There must lie a vector v_{r+1} that completes the span of V. We see that $v_1, v_2, \ldots, v_r$ doesn't span V. We are also assuming that adding v_{r+1} to the system is also independent. So, let's check this linear indpendence by setting $c_1 v_1 + \ldots + c_r v_r + c_{r+1} v_{r+1} = 0$. Manipulating this expression gives us $v_{r+1} = -\frac{1}{c_{r+1}}(c_1 v_1 + \ldots + c_r v_r)$. This shows that v_{r+1} is in the span of the collection, which is contradictory to our supposition. Therefore, c_{r+1} must be equal to zero. This gives us $c_1 v_1 + c_2 v_2 + \ldots + c_r v_r = 0$, which implies that $\sum_{i=1}^{r} |c_i| = 0$, showing linear independence. □

2.6. Is it possible that vectors v_1, v_2, v_3 are linearly dependent, but the vectors $w_1 = v_1 + v_2$, $w_2 = v_2 + v_3$, $w_3 = v_3 + v_1$ are linearly independent?

Proof. This is immediately false since $v_2 = (w_1 + w_2 - w_3)/2$. Using that result, you can get every other vector v_i. □